泌尿外科系外科操作手冊

在診斷與治療過程中須確定維護病患的隱私與尊嚴

教學宗旨及目標

一. **宗旨**:醫學教育是泌尿科的重任，對於見習、實習、住院醫師輪調至泌尿科，吾人認為教育之宗旨乃在於

(1)傳授泌尿科基本知識與技能

(2)提供良好的臨床教學環境作為實、見習醫師進入臨床醫學的橋樑

(3)建立以病患為中心之醫療服務

(4) 針對泌尿學科從事有系統的回顧與複習

(5)協助見實習醫師自我認知，分析性向與生涯規劃。

二. **目標**：

(1) 基本泌尿疾病之診斷、處理及照顧

(2) 泌尿科急症之鑑別診斷與處理

(3) 泌尿科併發症之處理

(4) 感染與抗生素使用原理

(5) 醫學倫理與紀律

(6) 正確病歷寫作與理學檢查

(7) 國家醫師資格考試之重點

學習項目及範圍

1. 疾病的診斷、治療，包括尿路結石、尿路感染、尿路先天性異常、泌尿道腫瘤學、婦女泌尿學、男性學、排尿異常及失禁、小兒泌尿學泌尿內視鏡手術，腎臟移植及高齡泌尿學等。
2. 學習一般泌尿手術之方法及其手術之適應症，術後的照顧原則和併發症的處理。
3. 基本泌尿科檢查之判讀，包括尿液之分析與培養，KUB、IVP、超音波、電腦斷層、MRI，以及膀胱尿動力學檢查。
4. 研習體外震波碎石機的原理，並至震波室實際了解其操作過程。
5. 研習膀胱尿動力檢查之原理及其判讀和簡易操作過程。
6. 腎移植學。
7. 男性性功能異常之各項檢查項目之原理及其判讀和操作過程。

URO ORIENTATION:

一、工作職責

每位 Resident & Intern 在病房及開刀房方面，跟隨指定主治醫師為主，
必要時，總醫師得以分配於開刀房與病房的學習處理其它主治醫師之病例。

A. 病房事項

1) Resident，Intern 及 Clerk 在早上 7:00 VS or CR 查房前，先自行訪視病人(負責病床區為所跟 VS 的所有床位)及收集 Data (如 I&O、CXR、CBC、Biochemistry)同時應熟悉病人現在情況及用藥.全病房 Invasive procedure (如 on Foley, on CVP, endotracheal intubation and chest tube insertion, suprapubic cystostomy)，由 Resident 執行。
2) Resident & Intern 負責病房時，除了一般的外科 history taking & P.E. 外，在 review of system & P.E.要特別加入 Genitourinary system，including genital organs 及 Digital rectal examination 等之檢查註記。
3) 每天早上查房時間須視各 VS 而有不同。
4) 每天下班前如有特殊檢查或情況的病人，須與小夜 NSP & duty Resident or Intern 交班。
5) 與 NSP 配合，處理 order、Change dressing、拆線、拔除引流管等。
6) 至少每天紀錄一次 Progressive Note， chart round 前需熟悉病人情況及治療計畫並於 chart round 中簡報及討論。
7) Resident & Intern & NSP 須負責 key in OP schedule。
8) 請勿隨意拔除病人的 Foley catheter，Ureter catheter，& PCN tube，如須拔除者，一定須經過 CR or VS 同意。
9) Remove Foley 後，請將 Urine bag 內的尿液倒掉後用塑膠袋裝好，丟入污物桶;請記得 3-way Foley catheter 內的 Balloon 約 20-30cc.以上的蒸餾水，請完全抽掉後再拔除。
10) TUR-P 的病人術後如需 Release Foley catheter 的 traction,請在 Release traction 後,確實做好 Foley catheter care。
11) Pre-op order 需註明手術名稱及日期。
12) 值班時,如果護士通知某位病人的 Lab data, U/A 有 Pyuria (WBC > 5/HPF；或>30/uL)的情形,請 Order U/C 及給予經驗性抗生素 Cefazolin (500mg/pc) 2pc iv q8h,並與值班住院醫師報告,及討論劑量是否需要調整。
13) 術前是否做 KUB 及 Chest PA view ,請依據住院通知單是否勾選此項檢查來做決定。

B. 開刀房事項

1) 手術前一晚上, 對隔日開刀病人做評估,寫 Pre OP assessment ,由負責的 Resident or Intern 紀錄。

- Pre OP assessment Form:
  a) Diagnosis
  b) Chief complaint
  c) Past history
  d) PE (Positive finding)，including GU system, Gigital rectal examination
  e) LAB data. EKG, echo, and imaging study
  f) Op Indication
  g) Plan

2) 8:30 前入開刀房確定病人資料及手術方式及部位，,學習無菌觀念、正確刷手、導尿、消毒、鋪單、跟刀用心投入,積極發問。手術後幫忙傷口清潔及包紮,送病人至 POR。

3) 手術房 6 房、7 房、8 房、16 房、17 房 (醫學大樓 2F), 81 房 82 房 (兒童醫院 8L) 。

→Local anesthesia room 6 房(W1-5)開放 Intern 及 Resident 跟隨 Senior resident 學習 circumcision 及 Vasectomy

→General or Spine anesthesia room 7 房、8 房、16 房、17 房、81 房 (W1-6)

4) 非經 CR 許可, 除非當天手術已全部結束,不得自己先行離開開刀房。

5) CR 得以視 OR 之需要,調配 Resident & Intern 至 OR 支援。

二、 見習醫師工作內容說明：

見習工作以跟隨主治醫師見習一般住院病患的照顧及研習各疾病之診斷、治療為主。其工作地點包括病房、手術房、門診。

地點	見習工作項目	頻率	說　　明	負責醫師
病房	病房巡診	每日一次	參與查房巡診與床邊教學	主治醫師
	病歷書寫或心得	乙篇	記錄住院病人病情及處置	
手術房	見習各類手術之準備與施行	每週兩次	觀察各類手術之施行以及基本解剖的認識	
門診	門診教學跟診	每週一至兩次	由主治醫師在門診講授常見的門診疾病及診斷	

三、 實習醫師工作內容說明:

臨床工作以協助主治醫師及住院醫師,學習一般住院病患的照顧及研習各疾病之診斷、治療為主。其工作地點包括病房、手術房、門診及值班。

地點	實習工作項目	頻率	說　　明	負責醫師
病房	接新病人	每天與住院醫師配合	與住院總醫師或住院醫師一同接病人	總醫師及主治醫師
	病房巡診	每日一次	參與查房巡診與床邊教學	
	病歷寫作	每日一次	記錄住院病人病情及處置至少每天一次	
手術房	協助手術之 準備與進行	每天與住院醫師配合	實際觀察及了解各類手術之施行情形以及基本解剖的認識	主治醫師
門診	門診教學	每週兩次	由主治醫師在門診教學	主治醫師
值班	值班教學	每三天一次為原則	與住院醫師一起學習住院病人的照顧參與每日夜間病房巡查,參與總醫師夜間查房。	總醫師

四、 住院醫師工作內容說明:

臨床工作以協助主治醫師,學習一般住院病患的照顧及研習各疾病之診斷、治療為主。其工作地點包括病房、手術房、門診及值班。

地點	實習工作項目	頻率	說　　明	負責醫師
病房	接新病人	每天與實習醫師配合	與住院總醫師或實習醫師一同接病人	總醫師及主治醫師
	病房巡診	每日一次	參與查房巡診與床邊教學	
	病歷寫作	每日一次	記錄住院病人病情及處置至少每天一次	
手術房	協助手術之準備與進行	每天與實習醫師配合	實際參與及了解各類手術之施行情形以及基本解剖的認識	主治醫師
門診	門診教學	每月一次排定之教學門診	由排定之主治醫師在教學門診指導	主治醫師
值班	值班教學	每三天一次	學習住院病人的照顧參與每日夜間病房巡查,參與總醫師夜間查房。	總醫師

五、科內學術活動

	學術活動		Time	Location
W1	外科部 Teaching round		7:00-8:00 AM	第一會議室
W2	泌尿腫瘤團隊會議		7:30-8:30 AM	13G 會議室
W3	第一週	科務會議	7:00-8:00 AM	13G 會議室
	第二週	專題討論	7:00-8:00 AM	13G 會議室
	第三週	泌尿科教育演講	7:00-8:00 AM	13G 會議室
	第四週	與 x 光科聯合討論會	7:00-8:00 AM	13G 會議室
W4	每週四	各次專科晨會或產品介紹	7:00-8:00 AM	13G 會議室
W5	第一週	Journal meeting	7:00-8:00 AM	13G 會議室
	第二週	外科部資深教授演講	7:00-8:00 AM	第一會議室
	第三週	泌尿科 Morbidity and mortality	7:00-8:00 AM	13G 會議室
	第四週	大外科 Morbidity and mortality	7:00-8:00 AM	第一會議室

※W2 由負責的 Chief Resident summary and review，最後由團隊主治醫師及負責人莊主任 Comment

※W3 clerk, intern, R1, R2 不需參加科務會議。

※W4 請準時參加各次專科晨會

※ 每月會安排 VS teaching round、chart round、主任 bedside teaching，在 13G 討論室，日期列在該月值班表，請 clerk leader、R1 leader 負責與該時段老師提早聯繫並通知同學。請輪流紀錄並收取簽到表，放置於 13G 護理站會議記錄盒。

※ 每月第二個 W4，泌尿科 mortality and morbidity conference，請各級醫師提早準備並與 CR 討論。

※ 每月最後一個 W3，泌尿科與 X 光科王俐人醫師，舉行聯合討論會。

常見之Q&A

1. 尿道下裂

尿道下裂是一種尿道行程異常的先天性疾病,其尿道開口可能會於會陰部至龜頭的中線上的任何一點。治療的方法為尿道整形術，大多數可一期完成，其時機最好在學齡前施行，以免影響小孩的心理發育。如果採用兩期尿道整形手術，第一及第二期之間要隔 6 個月以上，部分病例可採用門診方式手術。

2. 疝氣及陰囊水腫

小孩子的腹股溝疝氣及陰囊水腫可能單獨存在,也可能二者並存,其病因乃睪丸在胎兒期由腹腔內下降至陰囊的過程中 processus vaginals 未完全閉鎖,至腹腔內容物脫垂入腹股溝或陰囊內形成疝氣。疝氣在診斷出來後須儘早手術以防止腸子箝頓壞死的併發症,手術法是疝氣修補術。processus vaginals 的末端為 tunica vaginals 包在 testis 的外圍, 其內若有液體積聚就形成陰囊水腫。陰囊水腫在幾個月大的小孩可嘗試用注射空針抽吸,如果復發則為了防止或治療併發症的發生, 最好手術治療, 其方法是陰囊水腫切除。

3. 隱睪症

隱睪症乃胎兒其睪丸由腹腔內下降至陰囊的過程未完成, 以致睪丸停留在腎臟及陰囊之間的某一點。如果嬰兒在出生後被發現陰囊內摸不到睪丸,則應懷疑此症。約有半數的病人在出生不到一月內下降至正常陰囊內之位置其後自動下降之機會越來越小。隱睪症常與腹股溝疝氣並存。陰囊具有散熱作用, 使睪丸處於較體溫略低的環境,精蟲形成細胞對溫度很敏感,長期在高溫之環境下,將使這些細胞發生組織學的變化而形成能力損害,最後終至不孕症。所以最好在一歲左右手術治療,手術法是睪丸固定術。

4. 膀胱輸尿管尿液逆流

膀胱輸尿管尿液回流大部分是先天性的因素,原發性如膀胱與輸尿管開口抗逆功能不佳或異位性輸尿管開口所造成,也可能是次發性因素,如神經性膀胱或尿路下段阻塞造成尿道瓣膜,最常見的症狀是尿路感染反覆發生以致發熱、寒顫、腰痛,這表示腎盂腎炎的併發症已存在。另一個併發症是腎臟輸尿管水腫, 如不及早治療則腎實質漸漸破壞終致出現尿毒症及高血壓。確實的診斷可由排尿性膀胱尿道攝影術獲得,核醫攝影、膀胱鏡檢查或其他特殊檢查也可幫助診斷。

除抗生素治療併發之尿路或腎炎之外，輸尿管重建手術之根本治療，本科首創之微創抗逆流手術，治療率達 99%且部分病例以門診方式手術。

5. 先天性腎盂輸尿管阻塞

Ureteropelvic junction obstruction

先天性腎盂輸尿管阻塞乃腎盂輸尿管接合處之結構異常,以致腎盂排尿阻塞而造成腎盂擴大及腎水腫。最常見的症狀及徵象是腹部腫塊、腰痛、腹痛及嘔吐。確切的診斷可由腎臟超音波及核子醫學檢查做診斷。

治療的方法主要是腎盂整型術。

6. 前列腺肥大與前列腺癌(Benign Prostate hypertrophy，Prostate cancer)

前列腺肥大與前列腺癌都是壯年及老年男性常見疾病。二者的病因並不相同,但是兩種疾病可能同時存在。在兩種疾病都會逐漸腫大的腺體壓迫到尿道而出現種種阻塞性的症狀,如小便困難必須用力、尿流細小緩慢、小便遲疑、頻尿、尿急、小便斷斷續續、夜尿、小便末段滴尿,偶而也可能有小便疼痛急血尿的症狀，阻塞嚴重時造成尿液滯留而有解不出小便。除病史及理學檢查以外,前列腺肥大的診斷可由前列腺超音波檢查及膀胱鏡檢查幫助建立診斷。前列腺癌之診斷可由前列腺超音波檢查及前列腺癌抗原(PSA)來幫助診斷。但確切診斷須由外科標本的病理切片檢查來證實，再以腹部診斷攝影幫助做分斯診斷(staging)。

→前列腺肥大之手術治療大部分可採用經尿道切除術,或用開放性前列腺切除術。

→前列腺癌之治療因分期及惡性度不同而有不同的方法,最早期者可用經尿道切除術。

BPH Pre-op evaluation:

IPSS score:

- **Frequency (F)**
- **Urgency (U)**
- **Nocturia (N)**
- **Weak stream (W)**
- **Intermittency (I)**
- **Straining (S)**
- **Incomplete Emptying (E)**

Total score: 35

7. 腎結石

腎結石所造成的症狀及徵象，常見的有腫痛、 脊柱肋骨夾角區的敲痛，尿路感染的症狀 (如發燒、 寒顫)血尿等。實驗診斷的方法包括尿液分析、 生化分析、 腎臟輸尿管膀胱平面 X 光、 靜脈內腎盂攝影、 前行性或逆行幸腎盂攝影，腎臟超音波檢查等，治療的方法包括內科保守療法 (增加水分供應、 止痛、 控制感染、 尿液的酸化或鹼化等)，及外科兩大類。外科治療的方法有各種開放性外科手術，經皮腎造口碎石術 (PCNL)，及體外震波碎石術 (ESWL)等。

8. 輸尿管結石

輸尿管結石多由腎結石掉落入輸尿管而來。期症狀及徵候可能包括腰痛、下腹痛、脊柱肋骨夾角區的敲痛，噁心、噁吐、血尿、尿路感染的症狀（如發燒、 寒顫等 ）或膀胱刺激之症狀（ 頻尿、尿急 ）排尿疼痛等。診段方法類似前述腎結石之診斷法，治療方法包括內科療法(如前述 ）和外科療法，一般而言直徑小於 4mm 的遠端輸尿管結石約有 90%可排出，4-6mm 者有 50%可排出，大於 6mm 者則只有 20%有會自動排出。外科療法包括各種開放性外科手術，經皮腎造口碎石術（PCNL)、輸尿管腎鏡取石術或體外震波碎石術（ESWL)等。

9. 膀胱結石

膀胱結石可能是原發性的，也可能是尿路上段掉落入膀胱而來。期症狀包括小便遲疑、頻尿、排尿困難、血尿、反覆尿路感染，或排尿突然中斷併發外生殖器之急性放射痛。

診斷方法包跨 X 光檢查及膀胱鏡檢查。
尿道結石的症狀是急性尿液瀦流合併會陰部疼痛，診斷方法包括逆行性尿道攝影及膀胱尿道鏡檢查。

治療方法大部分可以經尿道取石術，少數巨大結石可用開放性手術。

10. 體外震波碎石

- 治療的適應症：
  Dornier Lithotripter，Model HN3 or Siemens Lithostar 體外震波碎石適用於泌尿道結石的病人，例如腎盞結石、腎盂結石及輸尿管結石。
- 禁忌
  a. 不適合作全身及腰椎麻醉的人。
  b. 不能暴露在輻射線下的病人,如孕婦。
  c. 裝有心律調整器的病人。
  d. 有遠端密尿道阻塞的病人必須先治療暢通。
  e. 心衰竭的病人 (理想控制後，仍宜 ESWL)
  f. 有凝血不正常的人。
- 副作用：
  Renal subcapsular hematoma。

血尿。

11. 腎臟移植

一、器官來源

A. 活體捐贈

1. 有血親關係者（五等親之血親：祖父母、父母、子女、兄弟姊妹、兄弟姊妹之子女）

2.無血親關係之活體捐贈（夫妻）

B. 腦死捐贈

二、活體捐腎者的條件

1.捐腎者必須瞭解手術失敗的可能性（包括捐腎者和受贈者）。

2.捐腎者必須沒有外來的壓力或情緒作用而強迫捐腎。

3.捐腎者必須沒有明顯精神疾病和情緒不穩等狀態。

本院親屬活體捐腎者的條件

1.年齡在 20 至 60 歲之間的健康成人。

2.兩個腎臟功能皆正常。

3.無高血壓、肺結核等慢性疾病。

4.自願捐出腎臟且動機純正。

＊＊＊ 手術前準備

活體捐贈

1. 組織配對及抗體檢查的再確定。

2. 抽血檢驗和 X 光檢查報告的再確定。

3. 活體捐腎同意書的填寫（需捐腎者及二等親兩人，共三人連署簽名）。

4. 常規的手術前準備（禁食、灌腸等的解釋及目的）

腦死捐贈

1.組織配對及抗體檢查的再確定。

2.抽血檢驗和 X 光檢查報告的再確定。

Including U/A, HBsAg, anti-HBV, anti-HCV, liver function test KUB, and renal echo

<u>參考文獻與書籍</u>

1. 教科書

General Urology 15th Edition Tanagho and McAninch

Compell's Urology 8th Edition Walsh. Retik. Stwney. Vaughan

2. 醫學雜誌

Journal of Urology

Urology

British Journal of Urology

European Urology

台灣泌尿醫學會雜誌

長庚醫學雜誌

Surgical Treatment Guideline

I. Hypospadias

More than 200 techniques have been created for hypospadias repair, the modified technique also continuously appear, but the choice should be depend the situation of deformity and ingenuity of surgeon appropriately. Restoration of normal function and appearance is the ultimate goals, the principle of correction including

1. Orthoplasty of penis with straightening of penis (Resection of chordee or dorsal plication)
2. Advancement of meatus to the tip of glans
3. Creation a neourethra lumen for defect

Preoperative evaluation

1. A thorough history including the prior genital surgery in detail and physical examination, the possibility of intersex (including Karyotypic analysis) are prerequisites for corrective plan.
2. CBC, D/C, PT, and chest x-ray are depending upon the patient's condition or requirement of anesthetic doctor.
3. Surgery and anesthesia consent from patient's family.
4. NPO 6-8 hours preoperatively.
5. D5S0.225 supply via IV start from the night before operation.

Postoperative care

1. The virtually most repair at our department is done on outpatient setting, except the severe or complicated hypospadias, so the simplified dressing is mandated.
2. Dressing is applied on patient who has been extensively dissected or bleeding is anticipated, otherwise, opened wound care and covering with a cup, since the prepuce is a vessels abounding tissue and infection is not usual.
3. Suprapubically urinary diversion in case of extensive repair, otherwise, urethral stenting with a No.5 or No.8 feeding tube is sufficient for urinary drainage.
4. Parenteral antibiotics are not routinely used, except the procedure with free flaps transposition.
5. Instruct patient's family to keep attention of preventing dislodgement of urethral

stent.

II. Vesicoureteral Reflux

In consideration of preventing renal damage from the reflux uropathy and cost-effectiveness, surgical antireflux still the main treatment for vesicoureteral reflux (VUR). The correction included intravesical technique (Politano and Leadbetter or Cohen procedure) and extravesical technique (Lich or Zaontz procedure), endoscopic injection of biologic or nonbiological material. The “Minimally-invasive antireflux technique” has been created by our department since 1996, and which was done on out patient basis.

Preoperative evaluation

1. History taking included voiding and bowel habit and physical examination to assess the possibility of neurogenic bladder or the associated abnormality.
2. Except voiding cystourethrography (VCUG) to evaluate the VUR, ultrasound is the first-line screening, intravenous urography or magnetic resonance image are required in case of suspicion with duplex collecting system.
3. Cystourethroscopy examination in case with suspicion of ureterocele, ectopic ureter or diverticulum before undergoing surgery.
4. Patient is encouraged to empty the bladder before sending to OR.

Postoperative care

1. Oral analgesic.
2. Oral antibiotics for patient with urinalysis of pyuria.
3. Ultrasound followup postoperatively for the transient obstruction of antireflux ureter after surgery.
4. Selective post-op VCUG is suggested 3 months after operation

III. Ureteropelvic Junction Obstruction

Dismembered pyeloplasty has been routinely used for correction of ureteropelvic junction (UPJ) obstruction. The operation involved resection of the redundant renal pelvis and the UPJ. We routinely stent the UPJ with internal catheter after the pyeloplasty and which enhance the success rate and the procedure was set in outpatient or one day hospital stay.

Preoperative evaluation

1. Indication of pyeloplasty for UPJ obstruction including, marked hydronephrosis, decrease of renal cortex, or decrease or renal function in affected kidney more than 10% compare to the contralateral side.
2. Initial ultrasound examination for the baseline assessment.
3. CBC, D/C, Chest X-ray are relatively check.

Postoperative evaluation

1. Check daily urine amount and abdominal condition to assure the patency of the UPJ.
2. Ultrasound evaluation postoperatively.
3. Remember to remove D-J stent after 6~8 weeks postoperatively.

IV. Undescended testis.

Orchiopexy is preferably performed when the child is about one year old. The hormone administration with human chorionic gonadotropin or a luteinizing hormone-releasing hormone are not convincing by the data. Out patient orchiopexy is amenable for most patients, and laparoscopic orchiopexy or exploration are necessary in undescended testes of high position in abdominal cavity or nonpalpable testes.

Preoperative evaluation

1. History of mother during pregnancy (included medication), physical examination and external genitalia examination.
2. Duplex ultrasound with 7 MHZ transducer is routinely used for evaluation
3. Laparoscopic exploration in case of no testicular stricture from ultrasound, CT or MRI.
4. Hormone or Karyotype analysis in case with bilateral undescended testes and bizarre genitalia.

Postoperative case

1. Oral antibiotics for prophylaxis of wound infection, neomycin ointment for scrotal wound of testicular reposition.
2. Instruct patient's family keep attention to the inguinal or scrotal hematoma after go home.

V. Renal Transplantation

Kidney failure may affect many of the organ systems:

1. Nervous system - confusion, seizure, coma
2. Lungs - fluid accumulation, pulmonary edema
3. Heart - cardiac arrest from increased blood potassium levels
4. Gastrointestinal - nausea, anorexia pancreatitis
5. Musculoskeletal - muscle weakness, spasm, edema
6. Blood - anemia, bleeding tendencies
7. Electrolytes - increased potassium, phosphorus and/or calcium

Indications

Evaluation for transplant involves a nephrologist, transplant surgeon, transplant coordinator, psychologist and social worker. Important factors discussed are the patient's physical and mental condition to withstand and understand the problems associated with transplantation. The patient must understand the concept of immunosuppression medication. This medication reduces the chance of rejection. Heart function is evaluated to be sure the patient can tolerate the surgery. Blood is tested for major histocompatibility (MHC) with other organs (called HLA genes) and reactivity of antibodies present in the patient to common antigens (called PRA - panel reactive antibodies). There are usually 6 HLA genes that are compared between a donor organ and recipient. A greater match between these usually signifies better compatibility. A higher PRA usually indicates longer waiting time for a compatible kidney. Once a patient is accepted as a candidate for transplantation, the patient is placed on an active waiting list.

Contraindications

1. Active infection and malignancy. If a malignancy is treated, a sufficient follow up time to rule out recurrence is necessary.
2. Severe medical problems such as advanced heart or lung disease
3. Age by itself is not usually a contraindication, but due to organ shortage our program limits the age indication between 3 to 65 years old.
4. Severe psychological problem, such as schizophrenia, poor compliance cases.

Preoperative evaluation:

1st visit:

填寫腎移植病人資料調查表
Blood typing，HLA，LAS，CXR，KUB），
理學檢查

2nd visit

SMA-12，Sugar，CBC，Platelet，VDRL，CMV-IgG， CMV-IgM，EBV， HBsAg (if positive check HBV-DNA)，HBeAg，Anti-HBc，Anti-HBs，Anti-HBe，Anti-HCV

□Abd echo if HBsAg or HCV positive

□Renal echo

腎移植術前衛教

臨危捐腎患者的評估

A：ICU Donor

1、評估病人之注意事項：

＃病史：（1）腦死原因：
（2）過去病史：
DM H/T CVA
Liver disease Heart disease
Renal disease SLE
OP history：
（3）其它

＃Vital sing：Coma Scale E（ ）V（ ）M（ ）
Temp： HR： BP：
Urine amount：
Shock：（+）（-）
Fever：（+）（-）
Infection： URI（+）（-）

Pneumonia（+）(-)

Sepsis（+）(-)

Dopamine（+）(-)

Dose：2amp in D5W 500cc run cc/hr

其它：

#Lab： ABO（ ）RH（+）(-)

VDRL（ ）TPHA（ ）HIV Ab（ ）

HBsAg（ ）If positive HBeAg（ ）

Anti HCV Ab（+）(-)

CMV:

Cr：

U/A：RBC（ ）WBC（ ）Protein（ ）

Sugar（ ）Bacteria（ ）

CXR：

KUB：

Echo：p.r.n.（ ）

4、Keep Donor Vital sign Keep SBP > 100mmHg

（1）FFP to keep oncotic pressure

（2）LR challenge （500cc 為一單位）

（3）使用 High dose Dopamine 之病人

a.若有尿液則儘量降低 Dopamine dose

b.若無尿則打 Lasix1Amp，若再無尿就觀察

（4）出現 DI 時 Hydration

DDAVP（0.025）nasal

（5）If Cr > 2 以上，check Cr g8h

B：受腎者(Recipient)

（1）詢問：

意願：(Y) or (N)

血型：(A)(B)(O)(AB)；RH（+）(-)

身高： _____cm 體重： _____kg 年齡： _____y/o

洗腎：H/D（ ）QW：_____X____Years Where：

CAPD（ ）

最近病史： URI（+）(-)

UTI（+）(-)

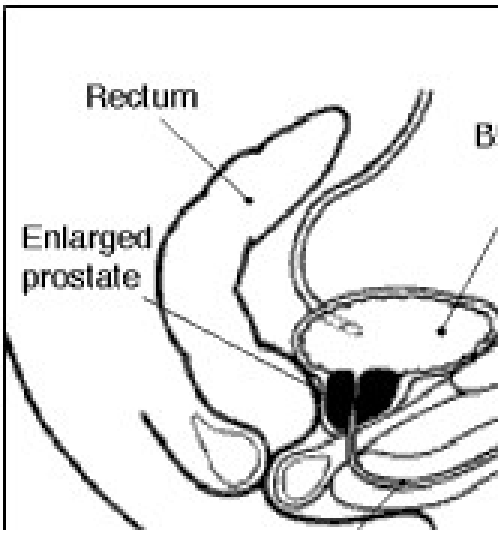
過去病史： Hypertension（+）(-)

DM（+）(-)
CV disease（+）(-)
Neuro disease（+）(-)
TB（+）(-)
Stone history：
Abdominal op history：
甲狀腺或副甲狀腺疾病：
B 肝（ ）C 肝（ ）

（2）、要求病人留行蹤或隨時可聯絡到的電話
（3）、告知病人 NPO

2、計算時間許可，應當天 H/D 之病人可繼續在外院 H/D，但須留電話以便聯絡
3、通知最後錄取之人選，請到 ER 掛號檢傷區報到
4、儘早聯絡洗腎室佔床，及與腎臟科 CR 聯絡洗腎時間

VI. Benign Prostatic Hyperplasia (BPH)



1. What is benign prostatic hyperplasia?

The prostate is a gland found between the bladder and the urethra .The prostate gland slowly grows bigger . Benign prostatic hyperplasia (BPH) is a condition that affects the prostate gland in men. As the prostate gets bigger, it may press on the urethra and cause the flow of urine to be slower and less forceful.

2. What are the symptoms of BPH?

Most symptoms of BPH start gradually.

- A hesitant stream
- Interrupted stream
- Weak stream
- More frequency
- Urgency and leaking
- Dribbling
- More frequent urination at night.

Many symptoms of BPH stem from obstruction of the urethra and gradual loss of bladder function, which results in incomplete emptying of the bladder. The symptoms of BPH vary, but the most common ones involve changes or problems with urination. The size of the prostate does not always determine how severe the obstruction or the symptoms will be. Some men with greatly enlarged glands have little obstruction and few symptoms while others, whose glands are less enlarged, have more blockage and greater problems.

Sometimes a man may not know he has any obstruction until he suddenly finds himself unable to urinate at all. This condition, called acute urinary retention, may be triggered by taking over-the-counter cold or allergy medicines. Such medicines contain a decongestant drug, known as a sympathomimetic. A potential side effect of this drug may be to prevent the bladder opening from relaxing and allowing urine to empty. When partial obstruction is present, urinary retention also can be brought on by alcohol, cold temperatures, or a long period of immobility.

It is important to tell your doctor about urinary problems such as those described above. In 8 out of 10 cases, these symptoms suggest BPH, but they also can signal other, more serious conditions that require prompt treatment. These conditions, including prostate cancer, can be ruled out only by a doctor's exam.

3. How will I know whether I have BPH or not?

Doctor will take a complete history of your symptoms, a digital rectal examination (DRE) is the next step. This DRE allows your doctor to feel the size of the prostate gland. The doctor needs to look at a sample of your urine for signs of infection. A blood sample will be checked for PSA. Other common tests may include: urine flow tests; ultrasound to measure urine left in the bladder and to check for bladder stones; urodynamic measurements using a catheter inserted into the bladder to measure the pressure of urine there; and transrectal ultrasonography

(TRUS) where an ultrasound probe is passed into the rectum to give a view of the prostate. A biopsy (samples of the prostate) may be collected using a needle to check for cancerous cells.

4. How will the doctor treat BPH?

Once the doctor is sure that your symptoms are caused by benign growth of the prostate gland, treatment will be recommended. However, your doctor may suggest that you wait to see if your symptoms get better on their own. If your symptoms get worse, doctor may suggest another treatment option.

Surgery is considered the most effective treatment and is used in men with strong symptoms. This is also the best way to diagnose and cure early cancer of the prostate. Surgery is usually done through the urethra, leaving no scars. Surgery does have risks, such as bleeding, infection or impotence. These risks are generally small.

5. Are there any drugs will be taken?

Drug treatments are available. Finasteride (brand name: Proscar) makes the prostate shrink, but it does not help all patients. The side effects of finasteride are rare and mild, but they usually have to do with sexual function. They go away when the medicine is stopped. The prostate will enlarge again after the medicine is stopped, so another treatment may have to be tried.

Another kind of medicine that was called alpha blockers also can help the symptoms of BPH. Some of these drugs are terazosin (brand name: Hytrin), alfuzosin (Xatral), doxazosin (brand name: Cardura) and tamsulosin (brand name: Harnalidge). Work by relaxing the muscles at the neck of the bladder and in the prostate. In this way they reduce the pressure on the urethra and so help increase the flow of urine. They do not cure BPH but help to alleviate some of the symptoms. The side effects of alpha blockers are mild and go away if you stop taking the medicine. The side effects include dizziness, fatigue and headiness.

6. Are there any surgery will be taken?

There are some main surgical options for BPH:

(1) TUR-P

Transurethral resection of the prostate (TURP) is the most common operation for BPH. The procedure is usually done under a spinal anesthetic. A long thin instrument

called a resectoscope is passed into the urethra. With a light source and lens on the end it acts as a telescope, allowing the surgeon to view the prostate either directly or on a video monitor. A precisely controlled electric current, applied by a loop of wire at the end of the resectoscope, is used to shave off sections of the enlarged prostate. See the separate BUPA fact sheet on TURP for further details. TURP is an effective procedure with over 90% of men reporting an improvement after the operation. However, as with any surgical procedure there is a risk of side-effects and complications. A common side-effect of this procedure is retrograde ejaculation - where semen passes into the bladder during orgasm instead of out of the penis. This is sometimes called a "dry orgasm". Retrograde ejaculation is usually not a problem, although it may reduce fertility. Complications of the operation can include urinary incontinence or damage to the urethra, resulting in a "stricture" that can itself cause difficulty passing urine.

(2) TUI-P

Transurethral incision of the prostate (TUIP) may be appropriate for men who have a less enlarged prostate. It is a quicker operation than a TURP and involves removing less tissue. It is performed under general or spinal anesthesia. As with a TURP an instrument is passed up through the penis, but instead of removing a portion of the prostate, small cuts are made in the neck of the bladder and the prostate. This reduces the obstruction of the flow of urine.

(3) Open prostatectomy

Open prostatectomy is only recommended for men whose prostate is very large. It is a major operation and carried out under a general anesthetic. An incision is made in the lower abdomen in order to remove the central part of the prostate.

(4) Other treatments

Laser therapy (using a laser probe to cut away prostate tissue) and transurethral microwave thermotherapy (using heat to remove some of the prostate tissue via a probe) are becoming more common in the treatment of BPH under special condition..

＊＊＊ 術前之準備:

A. 常規檢查包括血液、生化、尿液分析、胸部 x 光片、心電圖及凝血機能之檢查
B. 心、肺、肝、腎功能之評估,有必要時照會各專科醫師。
C. 有高血壓時要使用降血壓藥物控制，有服抗凝劑時需注意凝血功能。
D. 有糖尿病時控制血糖。
E. 有尿路感染或敗血症時,需做尿液及血液細菌培養，並使用抗生素。
F. 簽手術及麻醉同意書。
G. 手術前 8 小時進食 (前一天之午夜零時開始)。

H. 睡前需灌腸。

Ⅰ. 靜脈內點滴注射由前一晚或手術當天清晨開始。

J. 帶一瓶抗生素 (如頭孢黴素)至開刀房於手術前一小時給予。

＊＊＊ 術後之照顧:

A. 量記生命徵象。

B. 紀錄攝入及排出量。

C. 手術當天禁食，翌晨可開始進食。

D. 平躺 8 個小時(若以 spinal anesthesia)。

E. 靜脈內點滴液注射。

F. 必要時可打止痛針。

G. 抗生術注射。

H. 用生理時鹽水做連續性膀胱灌洗至手術次日。

I. 手術次日放鬆留置導尿管的牽引。

J. 要維持導尿管的通暢,如果導尿管的灌洗輸入量大於或等於輸出量時，要懷疑是否有阻塞的情形,可請醫師處理。

K. 如果有意識模糊不清或嚴重的血尿現象，應儘快請泌尿科專科醫師處理。

L. 留置導尿管要保持清潔並使用心黴素藥膏於尿道口附近,以免上行性感染。

M. 密切注意病人是否有 TUR-P syndrome 或 Spinal headache 之後遺症。

VII. Prostatic cancer

Abstract

The prostate is a gland in the male reproductive system located just below the bladder and in front of the rectum. It is about the size of a walnut and surrounds part of the urethra. The prostate gland produces fluid that makes up part of the semen.

Prostate cancer is found mainly in older men. As men age, the prostate may get bigger and block the urethra or bladder. This may cause difficulty in urination or can interfere with sexual function. The condition is called benign prostatic hyperplasia(BPH), and although it is not cancer, surgery may be needed to correct it. The symptoms of benign prostatic hyperplasia or of other problems in the prostate may be similar to symptoms for prostate cancer. Prostate cancer is the seventh most common cancer in men in Taiwan. The cancer develops from cells within the prostate gland. The majority of prostate cancers are slow growing and many men are unaware that they have this cancer. However, a small number of prostate cancers grow more quickly and may spread to other parts of the body.

Risk factors

- Age
  This cancer is rare in men under the age of 50 years.

- Family history
  The risk of developing prostate cancer increases if there is a first-degree relative (father or brother) who was diagnosed with prostate cancer at a young age. Having an elderly relative with prostate cancer is not uncommon and does not increase the risk. In a small number of cases, prostate cancer runs in families because of a faulty BRCA2 gene.

- Race

  Prostate cancer is more common in men of African descent.

Signs and symptoms

The prostate enlarges as men get older, and most men have some symptoms affecting urination. This very common condition, known as BPH, is not cancer. Many of the symptoms of prostate cancer are similar to those of BPH. Symptoms of prostate cancer include:

- urgently rush to the toilet to pass urine
- passing urine more often at night
- difficulty starting urine initially
- interrupted while passing urine
- incomplete emptied the bladder fully
- dribbling of urine
- blood in urine or semen
- pain in the back, hips or pelvis.

The above symptoms may not be caused by cancer, but men who notice any of them should consult their doctor for advice on treatment, and to rule out prostate cancer.

Screening

There is a blood test for a protein called PSA, which can help to detect prostate cancer. However, additional tests are necessary to be sure.

Detection

The following tests and procedures may be used:

- Digital rectal examination: The doctor inserted a lubricated gloved finger into the rectum and feels the prostate through the rectal wall for lumps or abnormal

areas.

- Prostate-specific antigen (PSA) test: A laboratory test that measures the levels of PSA in a blood sample. PSA is a substance produced by the prostate that may be found in an increased amount in the blood of men who have prostate cancer. PSA levels may also be high in men who have an infection or inflammation of the prostate, or benign prostatic hyperplasia (an enlarged, but non-cancerous, prostate).
- Transrectal ultrasound: A procedure used to examine the prostate. An instrument is inserted into the rectum, and sound waves bounce off the prostate. These sound waves create echoes, which a computer uses to create a picture called a sonogram. Transrectal ultrasound may also be used during a biopsy procedure.
- Biopsy: The removal of cells, tissues, or fluid to view under a microscope and check for signs of disease. There are 2 types of biopsy procedures used to diagnose prostate cancer:
    - Transrectal biopsy: A needle is inserted through the rectum into the prostate and a sample of prostate tissue is removed. This procedure is usually done using transrectal ultrasound to help guide the needle.

A pathologist will examine the sample to check for cancer cells and determine the Gleason score. The Gleason score ranges from 2-10 and describes how likely it is that a tumor will spread. The lower the number, the less likely the tumor is to spread.

Certain factors affect treatment options and prognosis (chance of recovery)

The treatment options and prognosis (chance of recovery) depend on the stage of the cancer (whether it affects part of the prostate, involves the whole prostate, or has spread to other places in the body), the Gleason score, the level of PSA, and the patient's age and general health.

used in the staging process:

- Radionuclide bone scan: A procedure used to create images of bones on a computer screen or on film. A small amount of radioactive material is injected into a blood vessel and travels through the bloodstream; it collects in the bones and is detected by a scanner.
- MRI (magnetic resonance imaging): A procedure in which a magnet linked to a computer is used to create detailed pictures of areas inside the body. This test is also called nuclear magnetic resonance imaging (NMRI).
- Pelvic lymphadenectomy: Lymph nodes in the pelvis may be removed to see if cancer has spread to them. Lymph nodes are small, bean-shaped structures that

are found throughout the body. They filter substances in fluid called lymph and help fight infection and disease.

- CT scan(CAT scan): A CT scan creates a series of detailed pictures of areas inside the body, taken from different angles. The pictures are created by a computer linked to an x-ray machine. This test is also called computed tomography, computerized tomography, or computerized axial tomography.
- Seminal vesicle biopsy: A needle is inserted into the seminal vesicles (glands that help produce semen) and fluid is removed and checked for cancer cells.
- Staging

  All cancers have "staging systems" that doctors use to describe the extent of the cancer or the degree to which it has progressed. The system used to stage prostate cancer has changed relatively recently, and so your doctor may use the old ABCD (Jewett-Whitmore) system to describe the stage of the disease or the currently recommended TNM system. Because this can get very confusing, we have given information from both these systems below.

  The **T stage** of the disease refers to the form of the primary **tumor** within the prostate. The **N stage** refers to the status of the lymph **nodes** near to the prostate (the pelvic lymph nodes). Finally, the **M stage** refers to the degree of **metastasis** -- the degree to which prostate cancer has traveled out of the immediate area of the prostate to other organs of the body.

Treatment

⑴ The Treatment of Localized Disease

Localized prostate cancer is cancer confined to the prostate, but can include situations in which the cancer invades into -- but not through -- the prostate capsule, which defines the border of the prostate. Such cancers are known as stage T1 or stage T2. Localized prostate cancer is *potentially curable* in a very high proportion of patients, which means that, all of the prostate cancer cells can be removed or killed, with the result that the patient does not go on to have recurrent prostate cancer.

There are three basic types of management currently considered to be standard options for patients with localized prostate cancer: (a) "watchful waiting," (b) radical surgery to remove the prostate, and (c) various forms of external beam radiotherapy and brachytherapy to kill prostate cancer cells. The use of "adjuvant" or "neoadjuvant" hormone therapies in combination with surgery or radiation.

These neoadjuvant hormone therapies for prostate cancer are controversial to a greater or lesser degree.

⑵ Treatment of Locally Advanced Disease:

There are several types of locally advanced prostate cancer, and we will take the time to itemize them here for clarity. This set of definitions will require a good appreciation of the "staging" systems used to discuss prostate cancer.

Unilateral and bilateral extracapsular extension

Prostate cancer can grow through the capsule and into the surrounding tissues. If this has happened from just one lobe of the prostrate, this is called *unilateral* extracapsular extension, which is classified as clinical stage T3/a/Nx/M0. By contrast, if this has happened from both lobes of the prostate, it is known as *bilateral* extracapsular extension and is classified as clinical stage T3b/Mx/M0.

Seminal vesicle-positive disease

It is common for prostate cancer to spread from the prostate into the seminal vesicles. Spread of prostate cancer from the confines of the prostate into the seminal vesicles is a second form of locally advanced prostate cancer. Regardless of whether the disease has spread into just one of the two seminal vesicles or into both seminal vesicles, it is classified as stage T3c/Nx/M0 disease.

Tumor fixed to or invading adjacent structures

If the tumor escapes the prostate and invades the bladder neck, or the external sphincter, or the rectum, then it is classified as clinical stage T4a/Nx/M0. Similarly, if the tumor escapes the prostate and invades the levator ani muscles or the pelvic wall itself, then it is classified as clinical stage T4b/Nx/M0.

Lymph-node positive disease

The final category of locally advanced prostate cancer is when the patient is found to have prostate cancer cells in his pelvic lymph nodes. In this case his disease is generally classified as *either* T3/N+/M0 disease *or* T4/N+/M0 disease, depending on whether he has T3 or T4 disease. In addition, the degree of lymph node invasion allows for subclassification as follows:

- N1 = metastasis in a single lymph node (2 cm or less in greatest dimension)
- N2 = either metastasis in a single lymph node (2-5 cm in greatest dimension) or multiple lymph node metastases (none more than 5 cm in greatest

dimension)

- N3 = metastasis in any lymph node greater than 5 cm in greatest dimension.

The general rule should be that the more advanced the disease, the less likely it is that purely surgical or purely radiotherapeutic interventions will succeed in controlling the disease. Indeed, there are now few surgeons who would consider that surgical intervention without some form of neoadjuvant or adjuvant therapy is appropriate for the management of most forms of locally advanced disease. Equally, the radio-oncologist now seems to accept that radiotherapy is likely to succeed in offering long-term progression-free survival if it is combined with hormonal manipulation of some form.

Basically, the list of *possible* categories of treatment is as follows:

A. radical retropubic prostatectomy or robot-assisted laparoscopic prostatectomy alone, which is probably only even potentially appropriate today in patients who are suspected of seminal vesicle involvement but in whom one or more attempts to biopsy the seminal vesicles were negative

B. radiation therapy alone, carried out using external beam radiation or brachytherapy or a combination of the two.

C. radical prostatectomy or robot-assisted laparoscopic prostatectomy followed by adjuvant (post-surgical)

hormone Therapy

D. Radiation therapy followed by adjuvant (post-surgical)

hormone Therapy

E. Radical prostatectomy or robot-assisted laparoscopic prostatectomy followed by adjuvant (post-surgical)

external beam radiation therapy

F. Neoadjuvant hormone therapy followed by radical

prostatectomy, with or without adjuvant (post-surgical)

hormone therapy

G. *Neoadjuvant hormone therapy followed by radiation therapy*,

With or without adjuvant (post-surgical) hormone therapy

H. *Neoadjuvant hormone therapy followed by cryotherapy*,

with or without adjuvant (post-surgical) hormone therapies.

⑶ The Treatment of Advanced Disease

- We consider "advanced" prostate cancer to be cancer in which there is any clear indication that cancer has escaped from the prostate such that there are *either* definitive signs or symptoms of metastatic disease (e.g., visible metastatic sites on a bone scan and/or prostate cancer-related bone pain) *or* clear indications of metastatic prostate cancer resulting from tests carried out with such prognostic indicators as PSA tests, RTPCR tests, immunoscintigraphy tests, and other developing molecular indicators of disease progression. We will continue to consider that a patient with positive lymph nodes (stage N+ disease) has a form of "locally advanced" disease. There are several different thought when it comes to the treatment of stage M+ prostate cancer. Implement a strategy of watchful waiting and delay treatment until symptoms appear.
- Treat advanced disease immediately with single modality hormonal therapy to minimize effects of testosterone. Major forms of such therapy in current use include
    - Orchiectomy (surgical castration) to remove the most important androgen-producing organs
    - Estrogen therapy to reduce LH production
    - LHRH agonist monotherapy used to reduce LH production
    - Antiandrogen monotherapy directed at the prostate and metastatic sites
- Treat advanced disease immediately with maximal androgen deprivation (MAD) or combined hormonal therapy (CHT) to minimize effects of testosterone *and* the adrenal androgens.

(4) Management of Hormone-Refractory Prostate Cancer

Regrettably, at this time, ***no form of hormone-refractory prostate cancer is curable***. All the available forms of therapy are *palliative*, which means that they can be used only to slow the progression of the disease and to relieve symptoms.

C. Radiation therapy

Radiation therapy is used extensively in the management of hormone-refractory prostate cancer in order to help manage pain associated with the growth of bone metastases.

D. Chemotherapy

Cytotoxic chemotherapy, one of the commonest forms of therapy for the majority of cancers, has been relatively unsuccessful in the treatment of hormone-refractory prostate cancer. Quite why this is the case we still do not know. However, it probably has something to do with the fact that, by comparison with other cancers, prostate cancer is a very slowly growing disease. Chemotherapy treatment is currently of limited value for prostate cancer. Doctors may use it if the cancer has spread outside the prostate gland and hormonal therapy has not worked, or has ceased to be effective. The aim of this treatment is to improve the quality of life.

VIII. ERECTILE DYSFUNCTION

Definition of erectile dysfunction (ED) (impotence):

"The consistent or recurrent inability to attain and/or maintain a penile erection sufficient for sexual performance." ***WHO-ISIR. 1st International Consultation***

Choosing ED therapies:

- First line

– **Education**
– **Treat underlying diseases**
– **Oral therapy (Apomorphine SL, PDE5 inhibitors, etc.)**
– **Androgen replacement therapy**
– **Cognitive & behavioral therapy**
– **Couple therapy**

- Second line

– **Combination oral therapy**
– **Vacuum device**
– **Transurethral or intracavernosal alprostadil**

- Third line

– **Penile prosthesis**
– **Vascular surgery**

Third-Line Therapy: Vascular Surgery

- Penile Revascularization
  only in carefully selected patients

– **Young men with traumatic pelvic fractures or congenital ED**

– **Contraindicated in men with diabetes, neurologic disorders, nicotine abuse**

- Venous ligation: only in some etiology proved patients poor long-term result

Third-Line Therapy: Penile Prostheses

- Intolerance or lack of response to other treatment modalities
- Irreparably damaged erectile tissue
- Specific concomitant medical conditions such as vascular or neurologic disease, chronic renal disease, and genital trauma (e.g., Peyronie's disease)

Inflatable Penile Prostheses: Reliability/Patient Satisfaction

- Long-term, US multicenter study AMS 700CX 3-piece inflatable penile prostheses
- 86% mechanical reliability after 5 y (n=372)
- 91% erection suitable for coitus (n=172)
- >85% would undergo surgery again and/or recommend procedure to a friend (n=178)

Carson CC, et al. *J Urol*. 2000;164:376-380.

Pre-operative evaluation:

1. History taken (sexual, medical, psychosocial, and surgical history etc.)
2. Physical examination included focused and neurologic examinations
3. Selected laboratory tests included lipid profile, HA1C, PSA, Testo and PRL etc.
4. Specialized testing for ED included diagnostic intracavernous injection, duplex Doppler ultrasonography of penile arteries, penile arteriography, dynamic infusion cavernosometry and cavernosography etc.

Post-operative care:

1. One night admission
2. Operative wound focused light compression dressing
3. On Foley catheter in patients with prosthesis implantation for one night
4. Antibiotic given
5. Analgesic given
6. Bed rest
7. Blood sugar control in DM patients

IX. Urolithiasis

1. Conservative observation

   Most Ureteral stones pass and do not require surgical treatment. Spontaneous passage depends on stone size, shape, and location. 90% of stones with size < 0.4cm will pass spontaneously. The majority of stones pass within 6 weeks after the onset of symptoms.

2. Relief of obstruction

   Obstructive urinary stone with fever and infected urine requires emergent drainage. Retrograde insertion of double-J ureteral stent is the first choice. If catheter cannot pass the obstructive stone, placement a percutaneous nephrostomy tube is necessary.

3. Extracorporeal shock wave lithotripsy (SWL)

   Ureteral stones fail to progress with conservative observation require further management. Renal stones less than 2.0-2.5cm in aggregate length are best treated with SWL. SWL is highly effective for uric acid and calcium oxalate dihydrate stones. Most stone fragments pass within 2 weeks. Cystine, calcium oxalate monohydrate and calcium phosphate dihydrate stones are relatively resistance to shock waves.

4. Ureterorenoscopy (URS)

   Ureterorenoscopy is effective for ureteral stone. Many instruments and methods have been developed, including stone basket extraction, electrohydraulic lithotripsy, lithoclast lithotripsy, ultrasound lithotripsy, and laser lithotripsy. Stone free rates range from 66 to 100% and are dependent on stone size and location, duration of stone impacted, and history of retroperitoneal surgery.

5. Percutaneous nephrolithotomy (PNL)

   Percutaneous removal of renal and proximal ureteral stones is the treatment of choice for large (>2.5cm) stones those resistant to SWL (suspected cystine or brushite stones), select low pole calyceal stones with narrow, long infundibulum and acute infundibulopelvic angle (infundibulopelvic angle <70°). First choice for staghorn stone is percutaneous nephrolithotomy. Residual stones can be managed with SWL or additional percutaneous procedures.

6. Open stone surgery

   Open surgery is the classic way to treat stone. However, the morbidity of open

surgery and the development of less invasive techniques have made open stone surgery relatively uncommon.

(1) Pyelolithotomy: Pyelolithotomy is effective for stone in renal pelvis, especially with an extrarenal pelvis.
(2) Anatrophic nephrolithotomy: Anatrophic nephrolithotomy is used for complex staghorn stone.
(3) Ureterolithotomy: Long-standing ureteral stones those inaccessible with ureterorenoscopy and those resistant to SWL can be removed with ureterolithotomy.

First Choice of Surgical Treatment for Urolithiasis

1. Renal stone
   Stone size < 1cm: SWL
   Stone size 1-2cm, Low pole: SWL or PNL
   All other locations: SWL
   Stone size > 2cm: PNL
   Staghorn stone: PNL; SWL or additional percutaneous procedures for residual stones.

2. Ureteral stone
   Stone size < 1cm, proximal ureter: SWL
   Stone size > 1cm, proximal ureter with mild to moderate hydronephrosis:
   SWL or URS
   Stone size < 1cm, distal ureter: SWL or URS
   Stone size > 1cm, distal ureter: SWL or URS
   Stone size > 1cm with severe hydronephrosis: URS, PNL, or ureterolithotomy

3. Bladder stone
   Standard treatment: cystolitholapaxy and cysto lithotripsy
   Relative large stones with narrow urethra in male: cystolithotomy

X. Bladder cancer

1. Tis
   → complete TUR-BT followed by intravesical instillation with BCG, Doxorubicin, Epidoxorubicin, Mitoxantrone or mitomycin.

2. Ta (single, low-to-moderate grade, not recurrent)
   → complete TUR-BT followed by bladder instillation.
3. Ta (large, multiple, high-grade, or recurrent)
   → complete TUR-BT followed by intravesical chemo- or immunotherapy
4. T1
   → complete TUR-BT followed by intravesical chemo- or immunotherapy
5. T2-4
   → radical cystectomy
   → neoadjuvant chemotherapy followed by radical cystectomy
   → neoadjuvant chemotherapy followed by concomitant chemotherapy and irradiation
6. Any T, N+, M+
   → systemic chemotherapy followed by selective surgery or irradiation.

Stress urinary incontinence

1. non-operative treatment :
   a. pelvic floor muscle exercise (Kegel exercise)
   b. biofeedback exercise : (utilization of electromechanical instruments to accurately measure, process and feed back to persons information about neuromuscular and autonomic activity both normal and abnormal in the form of analog or binary, auditory and /or visual feedback signals)
      → vaginal or anal device, electromyographic system
   c. electrical stimulation:
      → 12.5 Hz for urge incontinence.
      → 50 Hz for stress incontinence.
   d. assist instrument :
      → Introl (bladder neck support prosthesis)
      → Reliance Urinary Control Insert
      → Fem Assist (urethral occlusive device)
      → Fem-Soft
   e. Medical therapy :
      → Anticholinergics, oxybutynin chloride, flavoxate, imipramine, phenyl propanol amine, propantheline bromide, tolterodine, estrogen.
   f. Magnetic stimulation.
   g. Sacral nerve stimulation.
2. Operative treatment :
   a. Artificial urinary sphincter.

b. Bladder neck suspension :
   → Trans- abdomen : Burch procedure
   → Endoscopic : Stamey procedure
   → Trans- vagina : Kelly, Pereyra, Raz procedure
c. Urethral suspension :
   → Marshall-Marchetti-Krantz (MMK)
   → Urethral sling (TVT, SPARC)
d. bulking agents :
   → glutaraldehyde cross-linked (GAX) collagen.
   → micro-balloon implantation.

XI. Pheochromocytoma

Pre-op evaluation:

1. Urine VMA, Serum Catecholamine
2. MRI or MIBG scan

Pre-op preparation:

1. Bowel preparation: fleet enema 1 pc hs
2. Midnight NPO
3. CxR, EKG, CBC/DC, PT, APTT, electrolyte
4. Prophylactic parenteral antibiotic: cephalosporin to 1 vial to OR
5. Control Bp with Dibenzyline for two weeks.
6. Hydration by IVF for 1 day at least

Operation day:

1. On NG tube, on Foley catheter.
2. Prophylactic parenteral antibiotic: cephalosporin to 1 vial before surgery
3. CVP or Swan-Ganz for hemodynamic monitoring and intravenous vasodilator
4. On J-P drain, record J-P drain amount.

Post-op status:

1. Check F/S q6h * 1 day.
2. Check VS, CVP I/O q4h * 1 day.
3. Watch for hypotension and hypoglycemia.
4. Monitor J-P drain drainage amount for bleeding.